

M000 · Changing a Variable — English Worksheet (Intermediate)

Topic: Changing a Variable While You Build · **Course:** Builder Basics · **Level:** Intermediate · **Time:** about 38 minutes

1 · Predict 🧠

A variable can **change while the code runs**. `height = height + 1` means: take what `height` holds, add 1, put it back.

```
height = 1
repeat 4 times:
  place block down
  move up by height
  move forward
  height = height + 1
```

Does each step climb the same amount, or more each time?

Fill the table. Start at 1 and add 1 after every step.

Step	height before the step
1	1
2	
3	
4	

2 · Find the Bug 🐛

Each code block is broken. Write one or two sentences on what is wrong.

Bug A — Each step should climb higher than the one before.

```
height = 1
repeat 4 times:
  place block down
  move up by height
  move forward
```

What shape do you get instead, and what line is missing?

Bug B — The first step should climb 1, then 2, then 3.

```
height = 1
repeat 3 times:
  height = 1
  place block down
  move up by height
  move forward
  height = height + 1
```

Hint: look at the line inside the loop that sets `height` back to 1.

What is wrong?

3 · Fix the Code

The first step should climb **1**, then 2, then 3, then 4. Right now the first step climbs 2.

```
height = 1
repeat 4 times:
    height = height + 1
    place block down
    move up by height
    move forward
```

Write the fixed code:

Why does the order of those two lines matter? Write one sentence.

4 · Show Your Work 📷 🎥

Switch to your homework world. Build a **staircase** where each step climbs higher than the one before, using a variable that changes inside the loop. Then change **one number** — the start value or the amount you add — and run it again.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 • Your code. Scroll through it. Say what each part does. Point at the line that changes your variable.

2 · Your build.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.